

FINA2204 Tutorial 3: Hedging with futures

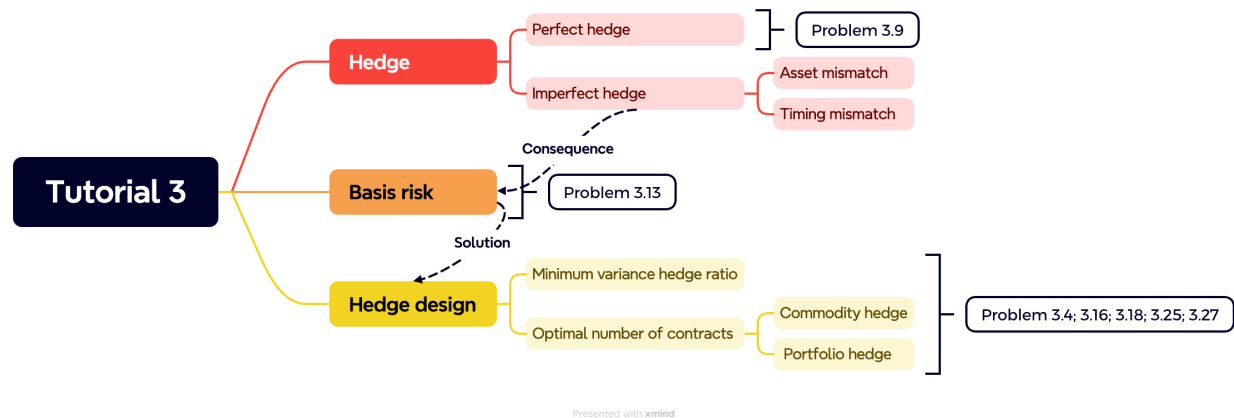
Zirui Song

UWA Business School

Aug 2026

Agenda

- Problem 3.4
- Problem 3.9
- Problem 3.13
- Problem 3.16
- Problem 3.18
- Problem 3.25
- Problem 3.27



Recap: perfect hedge

Key idea

Hedge in the direction that **offsets** your existing exposure: a **short hedge** (sell futures) if you own, or will own, the asset; a **long hedge** (buy futures) if you will need to purchase it.

Short hedge Suppliers

- WA wheat grower before harvest
- Gold miner with next quarter's production
- Exporter awaiting US-dollar payment

Long hedge Customers

- Airline buying jet fuel next quarter
- Food processor needing wheat in November
- Importer with a US-dollar invoice due

Q: When is a hedge “perfect”?

Recap: perfect hedge

- The gain/loss on futures **exactly offsets** the loss/gain in the spot market.
- This requires no mismatch in:
 1. Asset: futures and spot exposure refer to the same underlying asset;
 2. Timing: the hedge ends when the futures contract matures.

>> same asset + same maturity → perfect hedge

Recap: perfect hedge

Mechanism of convergence

1. Initial gap: spot and futures differ due to cost of carry (u), and the convenience yield (y): $F_0 = S_0 e^{(r+u-y)T}$.
2. Gradual narrowing: the price gap shrinks as the contract approaches expiration.
3. Convergence at maturity: Arbitrage forces perfect price alignment upon expiration. ($F_T = S_T$)

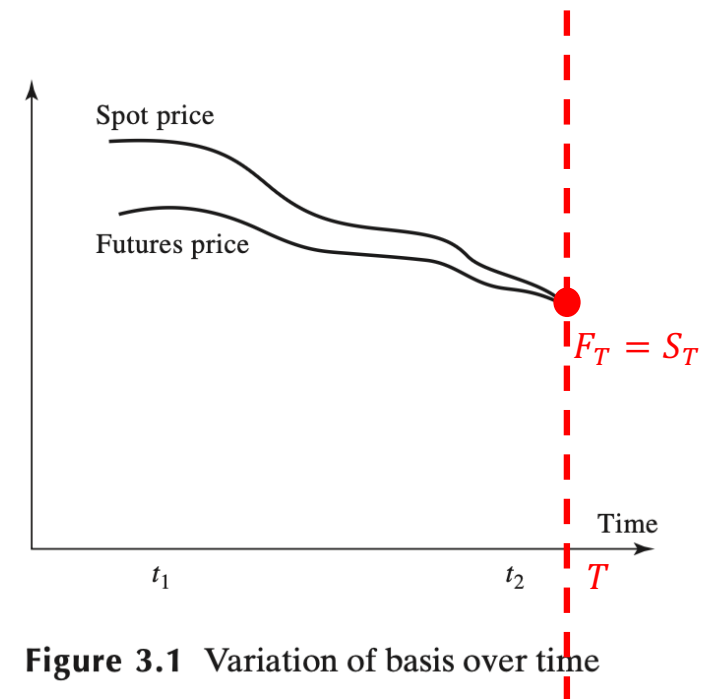


Figure 3.1 Variation of basis over time

Source: textbook, p.74

Recap: perfect hedge

Example 3.1 A short hedge

It is May 15. An oil producer has negotiated a contract to sell 1 million barrels of crude oil. The price in the sales contract is the spot price on August 15. Quotes:

~~Spot price of crude oil: \$60 per barrel~~

August oil futures price: \$59 per barrel $\rightarrow F_0 = 59$

The oil producer can hedge with the following transactions:

May 15: Short 1,000 August futures contracts on crude oil

August 15: Close out futures position

After gains or losses on the futures are taken into account, the price received by the company is close to \$59 per barrel.

$S_0 \rightarrow$ irrelevant

Convergence at maturity
 $S_T = F_T$

- What will be the outcome of this hedge if the spot price of crude oil on August 15 is (i) \$65 per barrel, (ii) \$55 per barrel?

Recap: perfect hedge

August spot price = \$65/bbl (so $F_T = \$65$)

Date	Spot	Futures
May 15	The \$60 spot price is not used!	Step 1: short futures contracts Sell 1,000 August futures @ \$59/bbl $SV = 1,000 \times 1,000 \times \$59 = \$59 \text{ million}$
Aug 15	Step 3: sell oil in the spot market Sell 1m bbls @ \$65 Revenue = \$65 million	Step 2: close out the futures position Buy 1,000 August futures @ \$65/bbl $SV = 1,000 \times 1,000 \times \$65 = \$65 \text{ million}$ Futures profit = -\$6 million
Net amount received (F_0) Effective revenue = \$65 million - \$6 million = \$59 million (= \$59/bbl)		

Price received for asset (S_2) Gain on futures ($F_2 - F_1$)

Short hedge locked in revenue of \$59/bbl

Recap: perfect hedge

August spot price = \$55/bbl (so $F_T = \$55$)

Date	Spot	Futures
May 15	The \$60 spot price is not used!	Sell 1,000 August futures @ \$59/bbl SV = 1,000 x 1,000 x \$59 = \$59 million
Aug 15	Sell 1m bbls @ \$55 Revenue = \$55 million	Buy 1,000 August futures @ \$55/bbl SV = 1,000 x 1,000 x \$55 = \$55 million Futures profit = \$4 million
	Effective revenue = \$55 million + \$4 million = \$59 million (= \$59/bbl)	

Short hedge locked in revenue of \$59/bbl.

Recap: perfect hedge

- The hedger did not lock in the May 15 spot price of \$60.
- Instead, it locked in the May 15 futures price of \$59, which reflects the market's forward valuation and cost-of-carry, not the current spot.
- **A perfect hedge locks in the initial futures price, not the initial spot price.**

Problem 3.9

- *Does a perfect hedge always succeed in locking in the current spot price of an asset for a future transaction? Explain your answer.*

Answer 3.9

- No. Consider, for example, the use of a forward (futures) contract to hedge a known cash inflow in a foreign currency. The forward (futures) contract locks in the forward exchange (futures) rate, which is in general different from the spot exchange rate.

Recap: basis risk

- The perfect hedge is too good to be true.
- Basis risk can arise when:
 1. Asset mismatch: the asset being hedged is not identical to the asset underlying the futures contract (cross hedging).
 2. Timing mismatch: the hedge ends before the futures contract matures, so spot and futures prices have not yet converged. This may occur because:
 - The transaction date is uncertain; or
 - No futures contract matures exactly when the hedge ends.

Recap: basis risk

- What is the ‘basis’ here?

$$\text{Basis} = \text{Spot price of asset to be hedged} - \text{Futures price of contract used}$$

- or

$$b_2 = S_2 - F_2$$

- Perfect hedge at maturity: $b_2 = b_T = 0$.
- But having “**no basis risk**” does not necessarily mean $b_2 = 0$; it simply means b_2 is known with certainty.

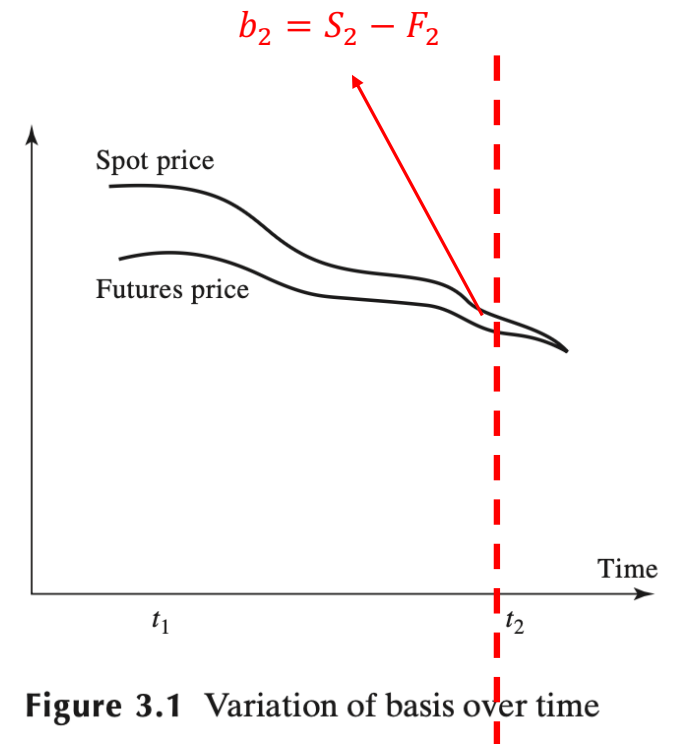
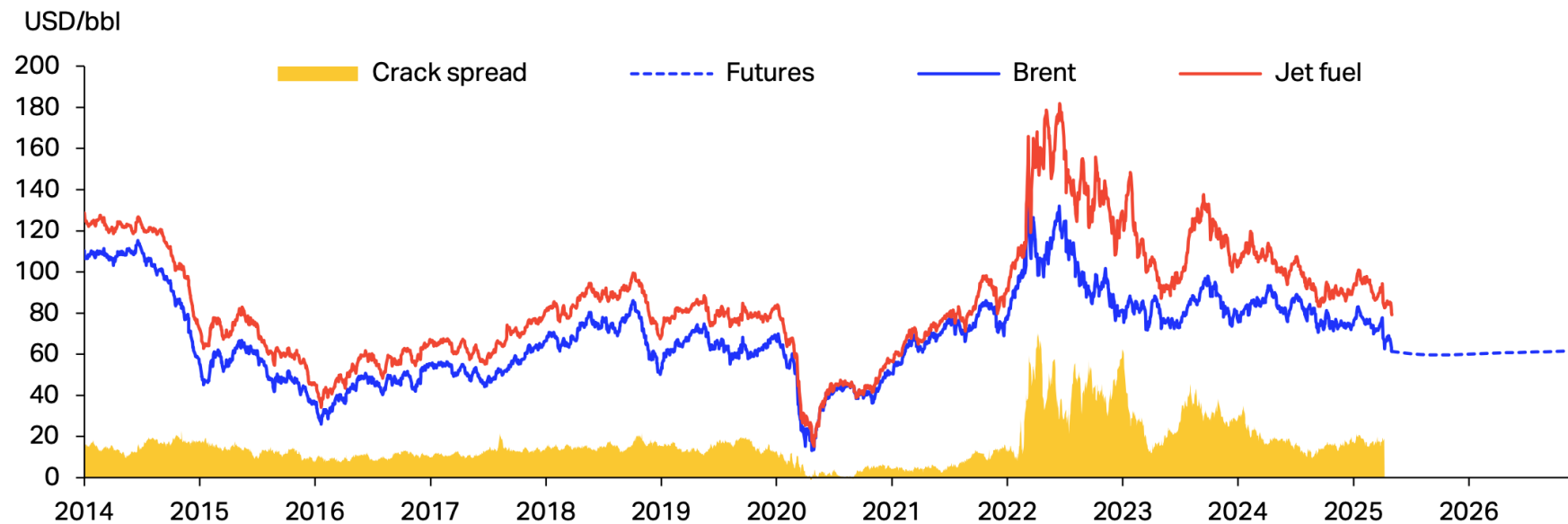


Figure 3.1 Variation of basis over time

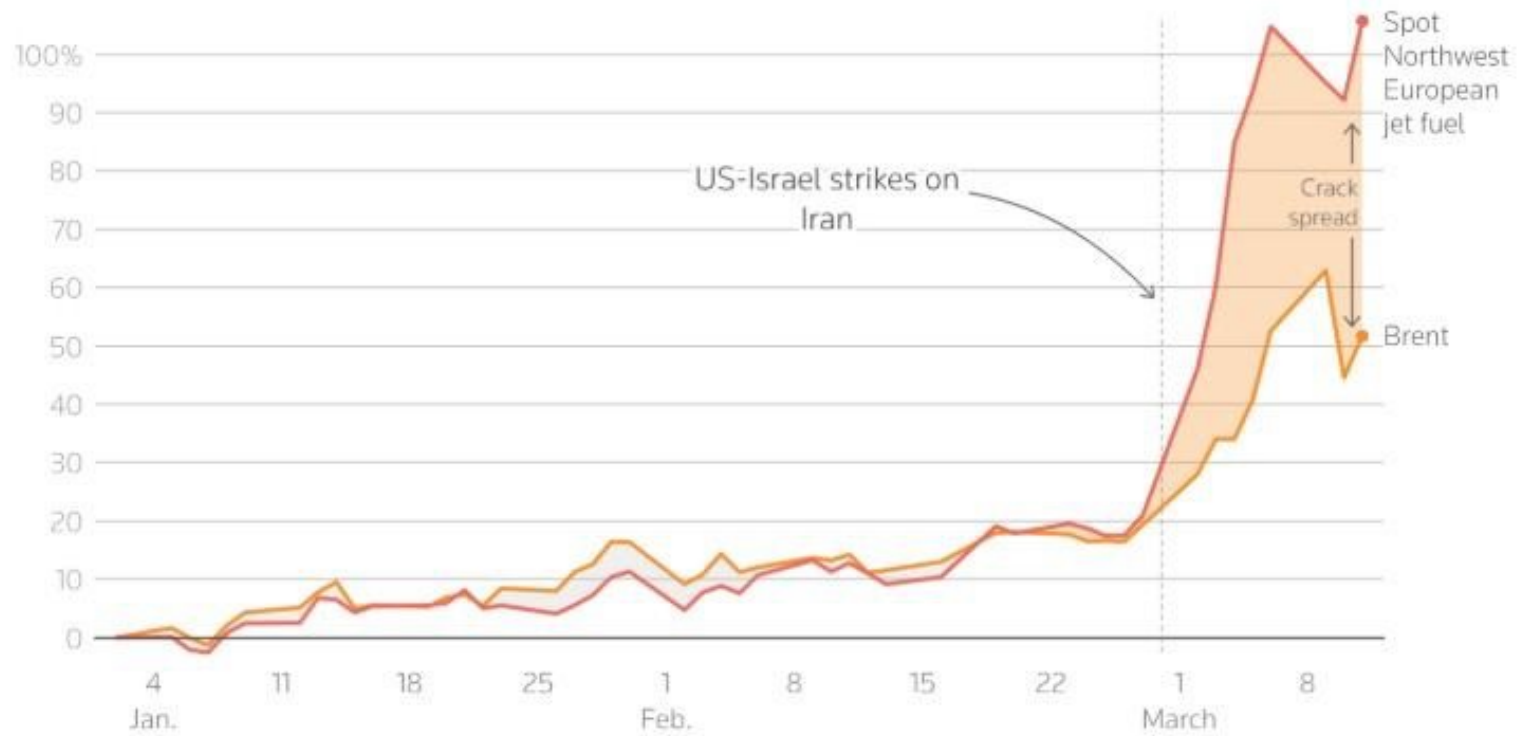
Recap: basis risk

Brent crude oil price with futures curve, jet fuel price, and jet crack spread, USD per barrel



Source: IATA Sustainability and Economics, Platts, ICE, Updated: 06/2025 Next Update: 12/2025

Recap: basis risk



Source: Reuters / LSEG.

Recap: basis risk

- Asset mismatch and timing mismatch mean that spot and futures prices may not move one-for-one.
- As a result, a simple 1:1 hedge may no longer be optimal.
- We therefore use the minimum variance hedge ratio (h^*) to determine the optimal futures exposure.
- We then use h^* to determine the optimal number of futures contracts (N^*).
- **Goal: minimize the basis risk from an imperfect hedge.**

Recap: hedge design

- Hedge ratio (h) is the proportion of the spot exposure hedged with futures.
- The optimal hedge (h^*) is the value of h that minimizes the **variance** of the hedged position.

$$h^* = \rho \times \frac{\sigma_S}{\sigma_F}$$

relative volatility

where:

- σ_S = **standard deviation** of % changes in spot price
- σ_F = **standard deviation** of % changes in futures price
- ρ = **correlation** between % changes in spot (S) and futures (F) prices

Non-examinable!

Recap: hedge design

- **Variance** measures the dispersion of a random variable around its expected value

$$Var(X) = \sigma_X^2 = E[(X - E[X])^2]$$

Special case: $Var(c) = 0!$

- **Standard deviation** is the square root of variance:

$$SD(X) = \sigma_X = \sqrt{Var(X)}$$

- In hedging, we use:

$$\sigma_S = SD(\Delta S), \sigma_F = SD(\Delta F)$$

to measure the **volatility** of spot and futures price changes.

Non-examinable!

Recap: hedge design

- **Covariance** measures how two random variables move together:

$$\text{Cov}(X, Y) = \sigma_{XY} = E[(X - E[X])(Y - E[Y])]$$

- $\text{Cov}(X, Y) > 0$: tend to move in the same direction
- $\text{Cov}(X, Y) < 0$: tend to move in opposite direction
- **Correlation** standardizes covariance into a unit-free measure:

$$\rho = \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y}$$
$$-1 \leq \rho \leq 1$$

- If $\rho = 0$, it means no linear correlation.

Non-examinable!

Recap: hedge design

Where $F_1 + b_2$ comes from

Hedge opened at t_1 and closed at t_2 . F_1, F_2 are the futures prices, S_2 the spot price at close, and $b_2 = S_2 - F_2$ the basis at close.

Short hedge (you will sell)	
Price received for asset	S_2
Gain on futures	$F_1 - F_2$
Net received	$F_1 + b_2$

Long hedge (you will buy)	
Price paid for asset	S_2
Gain on futures	$F_2 - F_1$
Net paid	$F_1 + b_2$

Effective price = $F_1 + (S_2 - F_2)$

$b_2 \rightarrow 0$ (perfect hedge)

$\rightarrow$ uncertainty (imperfect hedge)

Expanded effective price = $S_2 + h(F_1 - F_2)$ $\rightarrow$ If NO basis risk, $h^* = 1$

Goal $\rightarrow$ choose h to minimize $\text{Var}[S_2 + h(F_1 - F_2)]$

$\rightarrow \min_h \text{Var}[\Delta S - h\Delta F]$

Non-examinable!

Recap: hedge design

Variance properties:

$$Var(aX + bY) = a^2Var(X) + b^2Var(Y) + 2abCov(X, Y)$$


$$\begin{aligned}\rightarrow Var(\Delta S - h\Delta F) &= Var(\Delta S) + h^2Var(\Delta F) - 2hCov(\Delta S, \Delta F) \\ &= \sigma_S^2 + h^2\sigma_F^2 - 2h\sigma_{SF}\end{aligned}$$

- Differentiate with respect to h , and set the F.O.C to zero:

$$\begin{aligned}\frac{\partial Var}{\partial h} &= 2h\sigma_F^2 - 2\sigma_{SF} = 0 \\ h^* &= \frac{\sigma_{SF}}{\sigma_F^2} = \frac{\sigma_{SF}}{\sigma_F^2} \times \frac{\sigma_S}{\sigma_S} = \frac{\sigma_{SF}}{\sigma_S\sigma_F} \times \frac{\sigma_S}{\sigma_F}\end{aligned}$$

$\rightarrow \rho$

Non-examinable!

Recap: hedge design

- Consider the linear projection (OLS):

$$\Delta S = \alpha + \beta \Delta F + \varepsilon$$

$$\text{Goal} \rightarrow \min_{\alpha, \beta} E[\varepsilon^2] = \min_{\alpha, \beta} E[(\Delta S - \alpha - \beta \Delta F)^2]$$

- The intercept removes the mean ($\alpha^* = E(\Delta S) - \beta E(\Delta F)$)

$$E[(\Delta S - \beta \Delta F - (E(\Delta S) - \beta E(\Delta F)))^2] = \text{Var}(\Delta S - \beta \Delta F)$$

$$\rightarrow \min_{\beta} \text{Var}(\Delta S - \beta \Delta F)$$

- Hedging problem: $\rightarrow \min_h \text{Var}[\Delta S - h \Delta F]$

$$\beta = h^*$$

Non-examinable!

Recap: hedge design

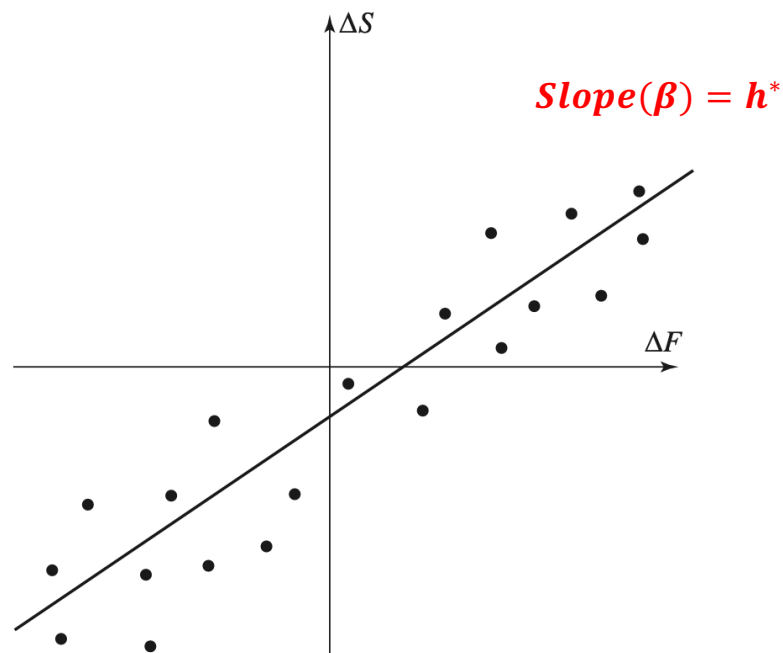


Figure 3.2 Regression of change in spot price against change in futures price

The hedge ratio h^* is the **slope** of the best-fit line from linear regression of ΔS against ΔF !

Non-examinable!

Recap: hedge design

Table 3.2 Data to calculate minimum variance hedge ratio when heating oil futures contract is used to hedge purchase of jet fuel

Month <i>i</i>	Change in heating oil futures price per gallon (ΔF)	Change in jet fuel price per gallon (ΔS)
1	0.021	0.029
2	0.035	0.020
3	-0.046	-0.044
4	0.001	0.008
5	0.044	0.026
6	-0.029	-0.019
7	-0.026	-0.010
8	-0.029	-0.007
9	0.048	0.043
10	-0.006	0.011
11	-0.036	-0.036
12	-0.011	-0.018
13	0.019	0.009
14	-0.027	-0.032
15	0.029	0.023

$$\overline{\Delta F} = \frac{0.021 + 0.035 + \dots + 0.029}{15} = -0.00087$$

$$\overline{\Delta S} = \frac{0.021 + 0.035 + \dots + 0.029}{15} = 0.0002$$

$$Var(\Delta F) = \frac{(0.021 + 0.00087)^2 + \dots + (0.029 + 0.00087)^2}{15 - 1} = 0.0009824$$

$$SD(\Delta F) = \sqrt{0.0009824} = 0.03134$$

$$Var(\Delta S) = 0.0006893$$

$$SD(\Delta S) = 0.02625$$

$$Cov(\Delta F, \Delta S) = \frac{(0.021 + 0.00087)(0.029 - 0.0002) + \dots + 0.00076397}{14} = 0.00076397$$

$$\rho = \frac{Cov}{SD \cdot SD} = \frac{0.00076397}{0.03134 \times 0.02625} = 0.9284$$

$$h^* = 0.9284 \times \frac{0.02625}{0.03134} = 0.778$$

Problem 3.13

- *‘If there is no basis risk, the minimum variance hedge ratio is always 1.0.’ Is this statement true? Explain your answer.*

Answer 3.13

- True.
- If the hedge is 1, the hedger locks in a price of $F_1 + b_2$.
- Since both F_1 and b_2 are known, this expression has a variance of zero and must be the best hedge.

Answer 3.13

$$\begin{aligned} \text{Goal} &\rightarrow \text{choose } h \text{ to minimize } \text{Var}[S_2 + h(F_1 - F_2)] \\ &\rightarrow \text{Var}(S_2 + hF_1 - hF_2) \\ &\rightarrow \text{Var}(S_2 + (hS_2 - hS_2) + hF_1 - hF_2) \\ &\rightarrow \text{Var}((1 - h)S_2 + h(F_2 - F_2) + hF_1) \\ &\rightarrow \text{Var}((1 - h)S_2 + hb_2 + hF_1) \end{aligned}$$

- Both F_1 and b_2 are known. The only uncertain is S_2 .
- So set $h = 1$, $(1 - h)S_2$ can cancel out, and the minimal variance is **zero** ($\text{Var}(b_2 + F_1) = 0$)!

Problem 3.4

- *Under what circumstances does a minimum variance hedge portfolio lead to no hedging at all?*

Answer 3.4

- No hedging: the proportion of the spot exposure hedged with futures is **zero**.

$$h^* = \rho \times \frac{\sigma_S}{\sigma_F} = 0$$

- Normally, $\sigma_S > 0, \sigma_F > 0$
- So the volatility ratio $(\frac{\sigma_S}{\sigma_F})$ is positive.
- Therefore, a minimum variance hedge leads to no hedging when the coefficient of correlation (ρ) is zero.

Answer 3.4

Variance properties:

$$Var(aX + bY) = a^2Var(X) + b^2Var(Y) + 2abCov(X, Y)$$

Goal $\rightarrow$ choose h to minimize $Var[S_2 + h(F_1 - F_2)]$
 $\rightarrow \min_h Var[\Delta S - h\Delta F]$

$$\begin{aligned} Var[\Delta S - h\Delta F] &= \sigma_S^2 + h^2\sigma_F^2 - 2h\sigma_{SF} \\ &= \sigma_S^2 + h^2\sigma_F^2 - 2h\rho\sigma_S\sigma_F \end{aligned}$$

1. The original
spot-price risk

2. The additional
risk from the
futures position

3. The risk reduction from the co-
movement between spot and future prices

- When $\rho = 0$, there is no hedging benefit. Any non-zero h only adds future risk, so $h^* = 0$; no hedge.

Recap: hedge design

For a commodity hedge:

- Risk is measured using **price changes**:

$$\text{Quantity } (Q) \times \text{Price change} = \text{Dollar P\&L}$$

- Let Q_A be size of spot position being hedged, Q_F be size of **one futures** contract.
- **Optimal number of futures contracts** (N^*) to use in a hedge:

$$N^* = h^* \frac{Q_A}{Q_F}$$

- N^* tells us how many futures contracts to use.

Problem 3.16

- *The standard deviation of monthly changes in the spot price of live cattle is (in cents per pound) 1.2. The standard deviation of monthly changes in the futures price of live cattle for the closest contract is 1.4. The correlation between the futures price changes and the spot price changes is 0.7. It is now October 15. A beef producer is committed to purchasing 200,000 pounds of live cattle on November 15. The producer wants to use the December live-cattle futures contracts to hedge its risk. Each contract is for the delivery of 40,000 pounds of cattle.*
- *What strategy should the beef producer follow?*

Answer 3.16

Setup

- Customer → fearing rising price → a long hedge
- Timing mismatch (underlying: Nov 15; futures: Dec)
- $\sigma_F = 1.4$
- $\rho = 0.7$
- $Q_A = 200,000$ pounds
- $Q_F = 40,000$ pounds
- $h^* = ? ; N^* = ?$

Answer 3.16

- The optimal long hedge ratio is

$$h^* = \rho \times \frac{\sigma_S}{\sigma_F} = 0.7 \times \frac{1.2}{1.4} = 0.6$$

- The beef producer requires **a long position** in $200,000 \times 0.6 = 120,000$ lbs of cattle. The beef producer should therefore take a long position in 3 December contracts closing out the position on November 15.

$$N^* = h^* \frac{Q_A}{Q_F} = 0.6 \times \frac{200,000}{40,000} = 3$$

Problem 3.25

- *A company wishes to hedge its exposure to a new fuel whose price changes have a 0.6 correlation with gasoline futures price changes. The company will lose \$1 million for each 1 cent increase in the price per gallon of the new fuel over the next three months. The new fuel's price change has a standard deviation that is 50% greater than price changes in gasoline futures prices.*
- *If gasoline futures are used to hedge the exposure what should the hedge ratio be?*
- *What is the company's exposure measured in gallons of the new fuel?*

Problem 3.25

- *What position measured in gallons should the company take in gasoline futures?*
- *How many gasoline futures contracts should be traded? Each futures contract is on 42,000 gallons.*

Answer 3.25

Setup

- Customer → fearing rising price → a long hedge
- Asset mismatch (the underlying: a new fuel; futures: gasoline)
- $\rho = 0.6$
- $\frac{\sigma_S}{\sigma_F} = 1.5$
- $Q_F = 42,000$ gallons
- $h^* = ?$; $Q_A = ?$; $N^* = ?$

Answer 3.25

- The hedge ratio (h^*) should be $0.6 \text{ (correlation)} \times 1.5 \text{ (relative volatility)} = 0.9$.
- The company has an exposure to the price of **100 million gallons (see the next slide for details)** of the new fuel. It should therefore take a position of 90 million gallons in gasoline futures ($\$0.9 \times 100,000,000$). Each futures contract is on 42,000 gallons. The number of contracts required is therefore

$$N^* = 0.9 \times \frac{100,000,000}{42,000} = 2142.9$$

- or, rounding to the nearest whole number, 2143.

Answer 3.25

The company will lose \$1 million for each 1 cent increase in the price per gallon of the new fuel over the next three months.

- Calculation Details for 100 million gallons
- Since it states that ‘the company will lose \$1 million for each 1 cent increase’
- If Q_A = exposure in gallons,
- And $\Delta P = 0.01$ (dollars per gallon),
- Then

$$\text{Quantity } (Q) \times \text{Price change} = \text{Dollar P\&L}$$

- Solving for Q :

$$Q_A = \frac{\text{Dollar P\&L}}{\Delta P} = \frac{\$1 \text{ million}}{0.01 \text{ dollars per gallon}} = 100 \text{ million gallons}$$

Recap: hedge design

For an equity portfolio:

- Risk is measured using **percentage returns**:

$$\text{Portfolio value } (V) \times \text{Return} = \text{Dollar P\&L}$$

- Let V_A be current value of the portfolio, V_F be current value of **one futures** contract
- Portfolio beta β measures the portfolio's **sensitivity** to market returns. ($\beta = h^*$)
- Optimal number of index futures contracts:

$$N^* = \beta \frac{V_A}{V_F}$$

Problem 3.18

- *On July 1, an investor holds 50,000 shares of a certain stock. The market price is \$30 per share. The investor is interested in hedging against movements in the market over the next month and decides to use the September Mini S&P 500 futures contract. The index is currently 1,500 and one contract is for delivery of \$50 times the index. The beta of the stock is 1.3.*
- *What strategy should the investor follow?*

Problem 3.18

Setup

- Investor / Supplier → fear falling price → a short hedge
- Asset mismatch (the underlying: a portfolio; futures: S&P 500)
- $\beta = 1.3$
- $V_A = 50,000 \times 30$
- $V_F = 1,500 \times 50$
- $N^* = ?$

Answer 3.18

- A **short** position in

$$N^* = \beta \frac{V_A}{V_F} = 1.3 \times \frac{50,000 \times 30}{50 \times 1,500} = 26$$

- contracts is required. It will be profitable if the stock outperforms the market in the sense that its return is greater than that predicted by the capital asset pricing model.

Recap: hedge design

- Index futures can be used not only to eliminate market risk but also to change the portfolio's beta.

Key idea

With β^* the **target** beta, write the subtraction so the bracket is *positive* and the direction reads off the words:

$$\text{to increase } \beta \rightarrow \beta^* : \quad \text{long} \quad N^* = (\beta^* - \beta) \frac{V_A}{V_F}$$

$$\text{to decrease } \beta \rightarrow \beta^* : \quad \text{short} \quad N^* = (\beta - \beta^*) \frac{V_A}{V_F}$$

$$\text{Market risk elimination: to decrease } \beta \rightarrow 0 : \text{short } N^* = (\beta - 0) \frac{V_A}{V_F}$$

- Buying index futures raises the portfolio's riskiness; selling them reduces it.

Problem 3.27

- *It is July 16. A company has a portfolio of stocks worth \$100 million. The beta of the portfolio is 1.2. The company would like to use the CME December futures contract on the S&P 500 to change the beta of the portfolio to 0.5 during the period July 16 to November 16. The index is currently 1,000, and each contract is on \$250 times the index.*
- *a) What position should the company take?*
- *b) Suppose that the company changes its mind and decides to increase the beta of the portfolio from 1.2 to 1.5. What position in futures contracts should it take?*

Answer 3.27

Setup

- Investor / Supplier → fear falling price → a short hedge
- Asset mismatch (the underlying: a portfolio; futures: S&P 500)
- Timing mismatch (the underlying: Nov 16; futures: Dec)
- Original $\beta = 1.2$
- $V_A = 100,000,000$
- $V_F = 1,000 \times 250$
- Target $\beta^* = 0.5, 1.5 \rightarrow N^* = ?$

Answer 3.27

a. The company should **short**

$$N^* = (\beta - \beta^*) \frac{V_A}{V_F} = (1.2 - 0.5) \times \frac{100,000,000}{1,000 \times 250}$$

or 280 contracts.

b. The company should take a **long** position in

$$N^* = (\beta^* - \beta) \frac{V_A}{V_F} = (1.5 - 1.2) \times \frac{100,000,000}{1,000 \times 250}$$

or 120 contracts.